Balance Rehabilitation Practical

Aims and Learning Outcomes

Aims

- To provide students with the opportunity to practice clinical and communication skills required for use in balance rehabilitation

Learning Outcomes

After the practical students should be able to

- Plan, implement and justify a balance rehabilitation programme
- Implement a rehabilitation plan to retrain postural alignment, sensory and movement strategies.
- Progress a rehabilitation plan to progress balance in siting and standing
- Teach a patient to adapt strategies to changing contexts.

Approaching balance rehab with patients

- Patient education to increase engagement
- Awareness of impairments and appropriate strategies to use
- Personalised Programme
- Setting GOALS - Use of outcome measures
- Addressing fear of falling - discussing risks and safety

Components of Balance Rehabilitation

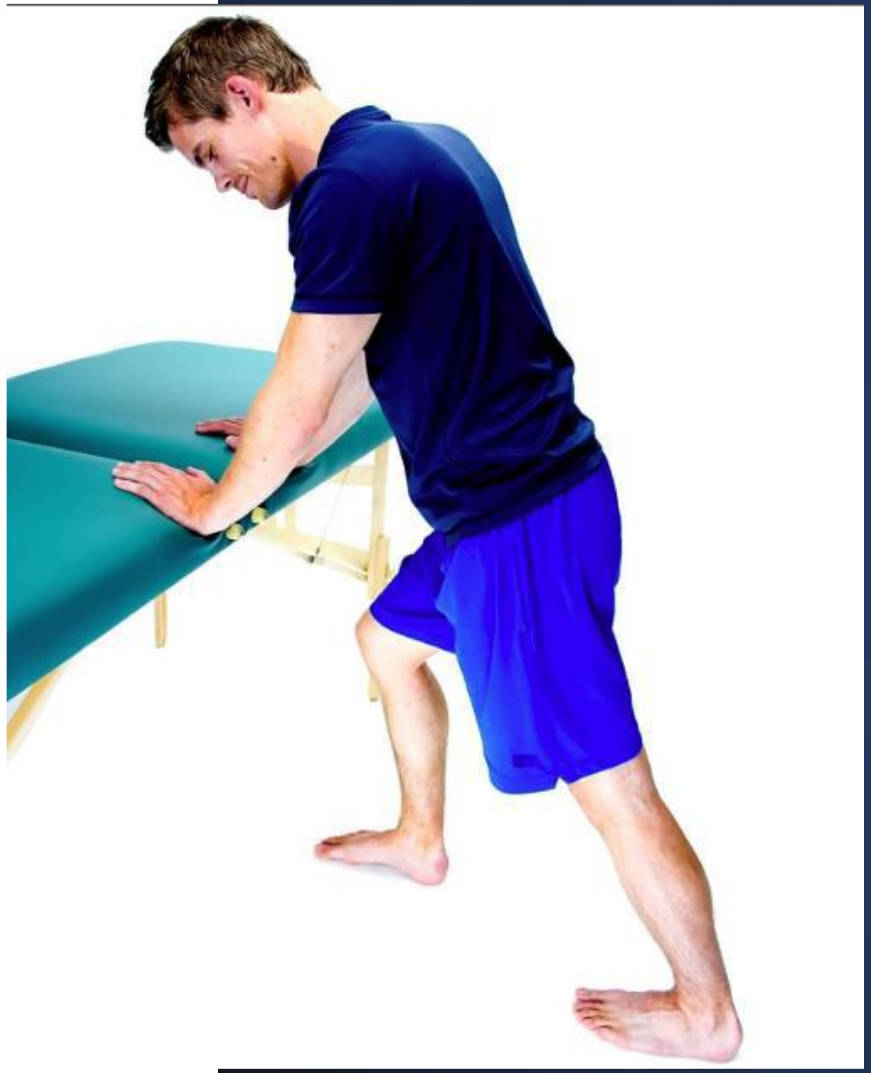
- Impairments
  - Treating musculoskeletal impairments; weakness and tightness

- Strategy
  - Retraining postural alignment
  - Retraining sensory strategies
  - Retraining movement strategies
  - Retraining anticipatory postural control
  - Cognitive strategies

- Function focussed exercises

Treating musculoskeletal impairments

- Muscle strengthening exercise
  - 3 sets of 10 reps
- Soft tissue stretches
  - * passively or self-stretches
- Ankle foot orthoses
- Night splints etc.
- Trunk exercises



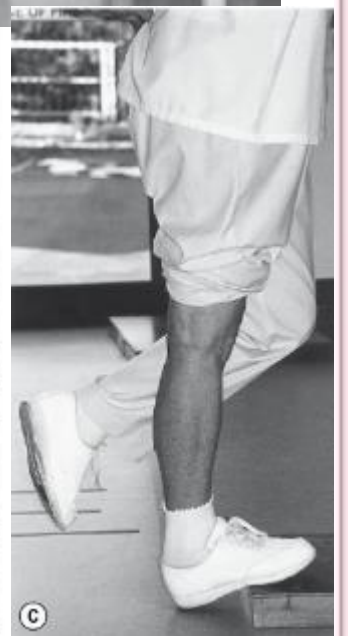
Examples of approaches to improve muscle strength











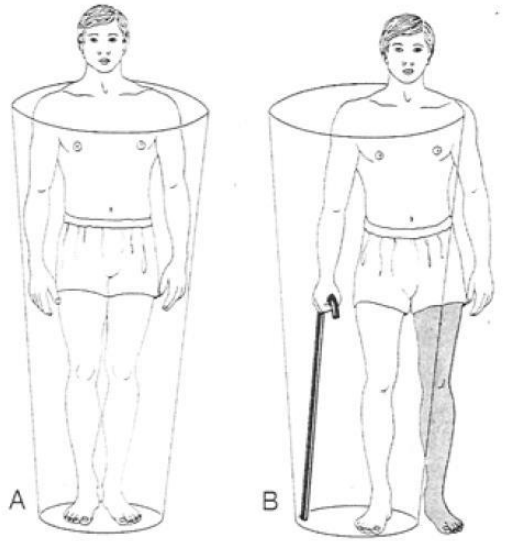
Box 2.1 Guidelines for functional strength training

- Exercises should be task-oriented – step-ups, squats, push-ups – and specific to tasks being learned plus standing up/sitting down, walking, stair climbing/descent, reaching to targets
- Encourage patient to exercise as intensively as possible, to the point of mild muscle fatigue*
- Grade the amount of resistance and number of repetitions to the individual's ability
- Utilize resistance from body weight, free weights, elastic bands, isokinetic dynamometry, exercise machines, inclined treadmill, depending on specific effects required and the existing muscle control and strength
- Vary body weight resistance by changing step height, chair height, weight of object being lifted
- As a general rule, a maximum number of repetitions should be performed (<10) without a rest and repeated in three sets, with a short break between sets
- In progressive resistance weight training, the person attempts between 3 and 20 repetitions of 50% of the maximum possible load that can be lifted in one attempt. The same principle for repetitions is followed for task-specific exercises, but using body weight for load
- For endurance, a higher number of repetitions is practised at low levels of load and exercises include stationary cycling, arm cycling and treadmill walking
- For very weak muscles, use methods which facilitate muscle activation and force generation, and include simple exercises, biofeedback, mental practice, functional electrical stimulation.
- A normal breathing pattern should be maintained throughout exercise

*Individuals at risk from high levels of blood pressure should avoid high-intensity resistance training.
Refer to American College of Sports Medicine's Guidelines for Exercise Testing and Prescription 2006.

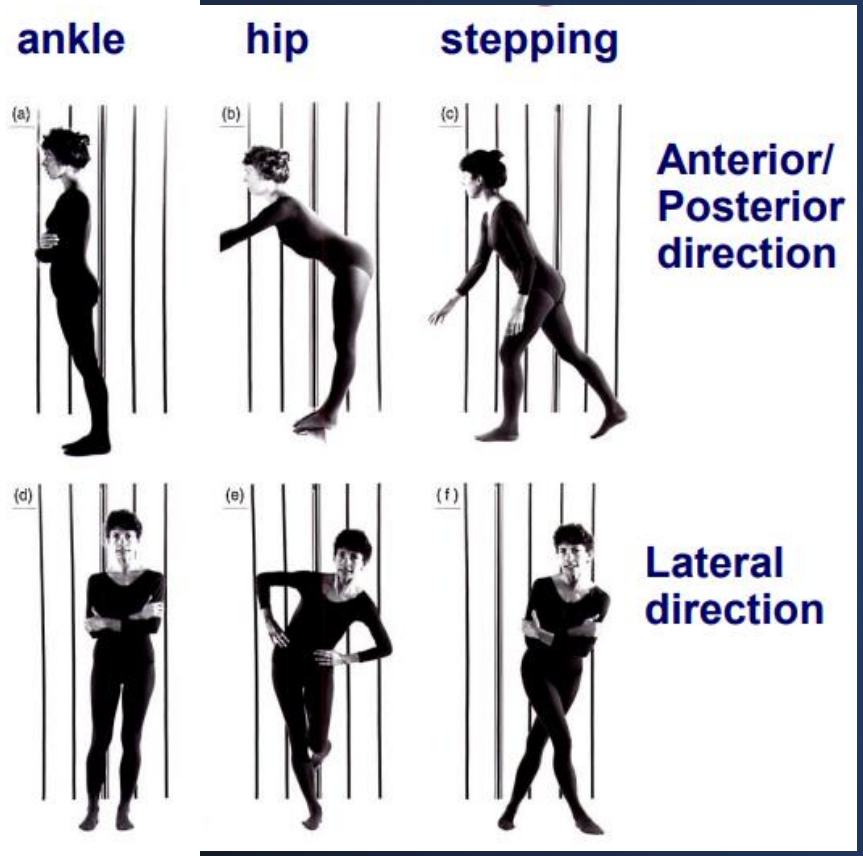
Strategy: Retraining postural alignment

- Verbal/manual cues to help patients find and maintain vertical posture
- Learning to maintain a vertical position with eyes open or closed, on firm or compliant surfaces.
  - Mirrors can provide visual feedback about position in space.
  - Ask the patient to line up the vertical stripe on his T-shirt with a vertical stripe on the mirror
  - Increase somatosensory cues concerning verticality: Patient leans against a small roll placed vertically on wall
  - Feedback can also be provided by using devices as simple as bathroom scales.



Retraining Reactive Balance

- In place strategies
  - Ankle
  - Hip
- Changing base of support strategies
  - Stepping strategies
- Train under varying conditions e.g. multiple directions, eyes open, eyes closed, compliant, incline surface, cognitive strategies



Retraining anticipatory balance

Example of movement include reaching, lifting, and throwing

Progression can be achieved by increasing:

- Task demands
- Speed, effort, degree of external support and task complexity





Retraining sensory strategies

- **Exercises for over-reliance on proprioceptive information:**
- Perform tasks while sitting or standing on surfaces with decreased somatosensory cues for balance: carpet, foam, pillows, moving surfaces-wobble boards.

- **Exercises for over-reliance on visual cues for balance:**
- Perform tasks where visual cues are absent (eyes closed, blindfolded) or reduced (blinders, diminished lighting).
- Visual cues can also be made inaccurate through the use of glasses smeared with petroleum jelly, prism glasses, moving curtains with stripes, moving visual scenes.

- **Exercises for enhanced ability to use vestibular information:**
- Exercises where one must maintain balance while both visual and somatosensory inputs are simultaneously reduced, such as standing on foam or a wobble board with eyes closed or moving visual scenes

Progressing balance exercises

- **Practice with sitting and standing**

- Increased time in static posture with eyes open and closed
- With reduced base of support (eyes open and closed)
- While changing the orientation of the head and trunk (e.g. looking over one's shoulder, leaning over)
- While performing a variety of upper extremity tasks beyond base of support
- As patient improves, more difficult and demanding tasks can be introduced.
- **Incorporating cognitive tasks:** e.g. maintaining standing balance with a small base of support while counting backwards in threes.
- **Dual attention:** e.g. walking while holding a glass of water; sitting on wobble board and trying to maintain balance while holding a glass of water with eyes open and eyes closed.

Functional exercises

- Example of exercises
  - Stepping to picking up objects beyond the person's limit of stability
  - Cross over stepping marks on the floor
  - Lunges in different directions
  - Stepping or dancing in time with music
  - Responding to external timing demands – catching a ball, kicking a ball, computer games
  - Coping with complexity in the environment
    - Negotiating an obstacle course
    - Making sudden stops and starts
    - Stepping over objects
    - Tandem walking
    - Marching on the spot with dual tasking
    - Walking and talking
    - Walking and carrying an object

